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Mendelian Randomization and Mediation Examination of the IgG N-Glycome Link Between Lifestyles and Age-Related Diseases

Haotian Wang¹  | Jingyu Wang²  | Xiaoni Meng³  | Shengzhi Sun¹  | Jinlong Li^{2,4}  | Youxin Wang^{2,5,6} 

¹School of Public Health, Capital Medical University, Beijing 100069, China | ²School of Public Health, North China University of Science and Technology, Tangshan 063210, China | ³Department of Clinical Epidemiology, Beijing Institute of Respiratory Medicine and Beijing Chao-Yang Hospital, Capital Medical University, Beijing 100020, China | ⁴Hebei Key Laboratory of Organ Fibrosis, Tangshan 063210, China | ⁵Nutrition and Health Innovation Research Institute and School of Medical and Health Sciences, Edith Cowan University, Joondalup, Western Australia, Australia | ⁶Tangshan Key Laboratory of Clinical Epidemiology, Tangshan 063210, China

Correspondence: Jinlong Li (lijinlong@ncst.edu.cn) | Youxin Wang (wangyouxin@ncst.edu.cn)

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ABSTRACT

Background: Lifestyles are associated with age-related diseases through chronic inflammation, yet the mediating pathways remain unclear. Immunoglobulin G (IgG) regulates inflammation by altering N-glycans. However, it lacks evidence on the mediating effect of IgG N-glycosylation linking lifestyles with age-related diseases.

Methods: A two-step Mendelian randomization (MR) design was conducted in East Asian and European populations to evaluate the mediating role of IgG N-glycosylation linking lifestyles with age-related diseases. In European populations, 32 lifestyles were selected as exposures, 23 glycan direct traits as candidate mediators, and 25 age-related diseases were considered in the MR analyses. For East Asian populations, 13 lifestyles, 24 candidate mediators, and 14 outcomes were analyzed. We used inverse-variance weighting as the primary method, supplemented by sensitivity analyses.

Results: Mediation MR analyses were conducted on three pathways in which IgG N-glycans were considered potential mediators among European ancestry. We observed the suggested mediating role of glycan peak (GP) 10 linking ground coffee consumption with rheumatoid arthritis (RA) ($\beta = -1.614$, $p = 0.023$), but not for GP15 ($\beta = 1.385$, $p = 0.069$). No mediating effect of GP22 was observed between dried fruit consumption and cognitive function ($\beta = 0.063$, $p = 0.092$). Additionally, fortified wine consumption showed no mediating effect on Alzheimer's disease via N-glycans, including those of GP7, GP19, GP23, and GP24 (all $ps > 0.05$). No mediating pathways were detected in East Asian populations.

Conclusion: The mediation MR analysis provides suggestive evidence that GP10 may play a role in the association between ground coffee consumption and RA. Further prospective cohort studies and mechanistic experiments are warranted to explore the potential underlying pathways.

Haotian Wang and Jingyu Wang contributed equally to this work.

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1 | Introduction

Aging is a complex process that leads to widespread functional decline and increased vulnerability to age-related diseases such as osteoarthritis, dementia, cardiovascular diseases, and cancer, ultimately leading to death. To prevent age-related diseases and delay aging, there is an urgent need to focus on the basic aging mechanisms underlying aging and these diseases. Previous studies have identified chronic inflammation as a hallmark of aging [1–3]. Notably, unhealthy lifestyle factors might promote systemic chronic inflammation, which in turn can lead to age-related diseases [3]. Moreover, geroscience has assumed that aging and age-related diseases share some basic mechanistic pillars that largely converge on inflammation [4, 5]. In particular, low-grade inflammation, known as “inflammaging,” develops during aging [3, 6, 7]. Geroscience has emphasized that these hallmarks do not operate separately but are interconnected. Hence, eliminating inflammation might serve as a potential approach for delaying age-related diseases and even antiaging.

Glycosylation acts as a universal posttranslational modification that profoundly affects protein function, stability, subcellular localization, and other traits [8, 9]. Immunoglobulin G (IgG), in particular, possesses two conserved N-glycans that are attached to asparagine 297 of the IgG CH₂ domain and act as a switch between the pro- and anti-inflammatory status of IgG [10]. Structural variations in IgG glycosylation have been implicated in the aging process and various age-related diseases [11–20]. In addition, IgG glycosylation is dynamic and can be altered by both lifestyle-based interventions and pharmacological interventions, while lifestyles are commonly associated with age-related diseases or their development. Observational studies have reported that there are anti-inflammatory alterations in IgG N-glycome patterns, which were observed from individuals participating in physical exercise [21], a low-calorie diet [22], and bariatric surgery [22].

Given that IgG N-glycosylation varies during the aging process as well as age-related diseases and can be modulated by lifestyle interventions, we hypothesized that the effect of lifestyles on age-related diseases might be mediated through the N-glycosylation of IgGs to their effectors (pro- and anti-inflammatory). However, investigations using cohorts to explore the putative mediating role of IgG N-glycosylation in the association between lifestyles and age-related diseases require a long follow-up period and a large sample size, for which evidence has been scarce. Mendelian randomization (MR) provides a framework that treats genetic variants as instrumental variables (IVs) to infer causal relationships, which benefits from their randomized assortment of genetic variants at conception; thus, causal associations are generally less susceptible to confounding bias and reverse causality [23, 24]. Recently, two-step MR, an extension of the MR framework, has allowed the estimation of the mediation effect [25, 26]. Here, we adopted a two-step design to detect the mediating effect of the IgG glycan profile between lifestyles and age-related diseases, as well as their mediated proportions.

2 | Methods

2.1 | MR Study Design

The present study leveraged the two-step MR method to identify the mediating effect of IgG N-glycosylation between lifestyles and

age-related diseases, as detailed in Figure 1A. In the MR design that utilized single-nucleotide polymorphisms (SNPs) as IVs, lifestyles were treated as exposures of interest, age-related diseases were outcomes, and IgG N-glycosylation was treated as a putative mediator. The causal interpretation of MR estimates relies on three assumptions of IVs: (1) the IVs termed SNPs should be strongly associated with exposures; (2) genetic IVs are not influenced by confounding variables; and (3) genetic IVs only indirectly affect outcomes via exposures. We estimated the *F*-statistic for each instrument SNP to assess its strength in conventional MR and the corresponding *R*² to quantify the proportion of phenotypic variance explained, with the former calculated as $F = R^2(N-2)/(1-R^2)$ and the latter as $R^2 = 2 \times \text{Effect}^2 \times \text{EAF}(1-\text{EAF})/[2 \times \text{Effect}^2 \times \text{EAF}(1-\text{EAF}) + \text{SE}^2 \times 2N \times \text{EAF}(1-\text{EAF})]$ or $R^2 = 2 \times (1-\text{EAF}) \times (1-\text{EAF}) \times \text{Effect}^2$, where EAF denotes Allele 1 frequency, SE the standard error, and *N* the sample size [27]. SNPs with *F*-statistics greater than 10 served as valid IVs to reduce the weak instrument bias in MR analysis [28].

In addition, this study adhered to the guidelines established by the Strengthening the Reporting of Observational Studies in Epidemiology Using MR (STROBE-MR) to ensure transparent reporting of MR research [29].

2.2 | Data Sources

2.2.1 | GWAS Data Sources for Lifestyles

The detailed information on GWAS data from lifestyles of interest used in the present study is presented in Table S1 and Figure 1B. We obtained summary-level data from the European publicly available GWASs of lifestyles, which included physical activity (moderate physical activity, moderate-vigorous physical activity, and vigorous physical activity) and 29 dietary habits. In addition, we downloaded summary-level statistical data related to 13 dietary habits of East Asian individuals from BioBank Japan (BBJ) via a link (<http://jenger.riken.jp/en/result>).

2.2.2 | GWAS Data Sources for Age-Related Diseases

We used publicly available GWAS summary statistics from European-ancestry populations, including data on rheumatoid arthritis (RA), stroke and its subtypes, osteoporosis, coronary artery disease, and myocardial infarction, 12 cancers, six neurodegenerative diseases, and five senescence-associated secretory phenotypes. We downloaded publicly available GWAS summary statistics of age-related diseases in East Asian populations from BBJ (<http://jenger.riken.jp/en/result>) for the following disease phenotypes: RA, ischemic stroke, osteoporosis, Parkinson's disease, and 10 cancers. Relevant information on the GWAS summary statistics is presented in additional Table S1.

2.2.3 | GWAS Data Sources for IgG N-Glycans

The present study takes IgG N-glycans as putative mediators between lifestyles and age-related diseases. We assessed the largest GWAS summary-level data of IgG N-glycans from a publicly available dataset that included 8090 European individuals [30], which provided GWAS data sources of 23 direct glycan traits (GP1-2 and GP4-24). For East Asian individuals, we collected blood samples from 536 Han Chinese individuals from Beijing

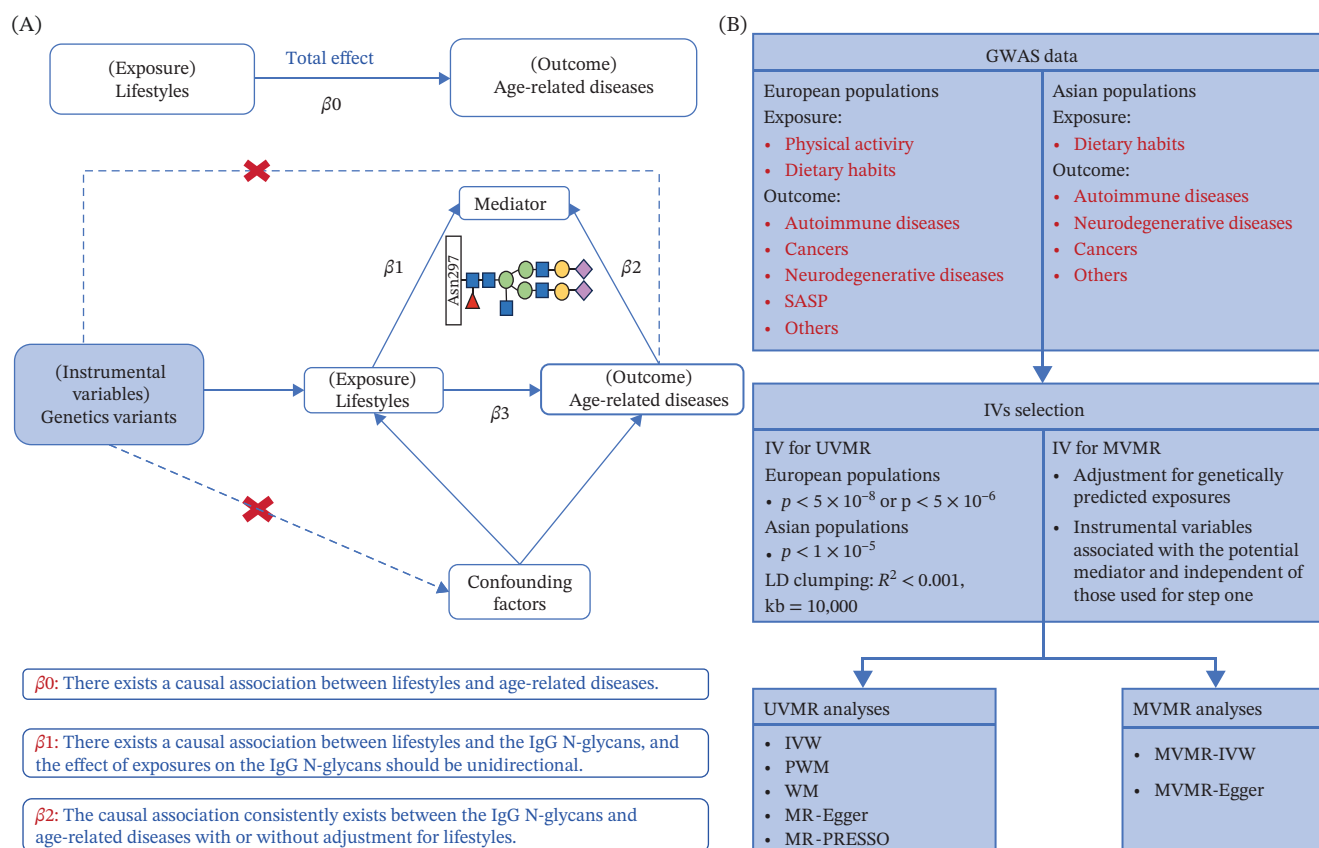


FIGURE 1 | Overview of the study design. IVW, inverse variance weighted; MR, Mendelian randomization; MR-PRESSO, Mendelian randomization pleiotropy residual sum and outlier; PWM, penalized weighted median; SASP, senescence-associated secretory phenotype; SNP, single-nucleotide polymorphism; WM, weighted median method. (A) The two-step MR design. (B) The analytical flow of the MR analysis.

Xuanwu Hospital and performed a genome-wide analysis. The detailed experimental process of IgG N-glycome was reported as described in the original articles [30, 31]. In brief, IgG was purified using protein G affinity chromatography, and glycans were released, labeled with 2-AB, and quantified by HILIC-UPLC, yielding 24 chromatographic peaks. Of note, GP3 was excluded from the European GWAS analysis as it coeluted with contaminants in the study samples. To minimize technical variation, raw glycan data were normalized by total area and batch-corrected using the ComBat function in the sva R package as detailed in the previous studies. The structures and descriptions of measured IgG N-glycan traits are presented in Table S2.

Mediators were determined by the causal associations between lifestyles and age-related diseases according to the following criteria: (1) causal associations existed between exposures and IgG N-glycans; (2) causal associations consistently existed between IgG N-glycans and outcomes with or without adjustment for exposures; and (3) the potential mediating effect of IgG N-glycans as mediators was identified in the context of significant associations between exposures and outcomes.

2.2.4 | IVs Selection

The selected genetic variants were identified according to the following steps. First, we selected SNPs that demonstrated a significant association with phenotypes at the genome-wide significance level ($p < 5 \times 10^{-8}$) in European populations; otherwise, we used

$p < 5 \times 10^{-6}$ and $p < 1 \times 10^{-5}$ for East Asian populations. Then, the SNPs were clumped using a linkage disequilibrium (LD) threshold of $r^2 < 0.001$ with reference to 10,000 kb, as defined by 1000 genomes from East Asian and European samples.

2.3 | Statistical Analysis

2.3.1 | Univariable MR (UVMR) and Multivariable MR (MVMR) Analyses

UVMR was implemented to estimate the causal associations of lifestyles with IgG N-glycosylation, with the inverse-variance weighted (IVW) method as the main analytical method [23], supplemented by the weighted median (WM) method [32], penalized WM (PWM) [32], and MR-pleiotropy residual sum and outlier (MR-PRESSO) [33] to enhance the robustness of causal inference. We also evaluated the horizontal pleiotropy of the IVs using MR-Egger regression. A p -value < 0.05 for the MR-Egger intercept was considered indicative of horizontal pleiotropy. Furthermore, the MR-PRESSO test was performed to detect and adjust for potential outlying variants. For MVMR, the MV-IVW and MV-Egger methods were applied to estimate the causal effect of IgG N-glycome on age-related disease in the condition of adjusting for lifestyles [34].

2.3.2 | Mediation Analyses

The mediation MR analyses were conducted in two steps [26]. In step one, we tested the causal associations of lifestyles with IgG

N-glycosylation using UVMR analyses (β_1). In step two, we utilized the framework of MVMR to assess the causal associations of IgG N-glycosylation with age-related diseases with adjustment for lifestyles (β_2). The product of the coefficients method was adopted to estimate the mediation effect of IgG N-glycosylation, that is, ($\beta_1 \times \beta_2$). The mediation proportion between lifestyles and age-related diseases that was mediated by potential mediators was calculated by dividing the mediation effect by the total effect. We used effect estimates obtained from MR analysis to derive standard errors using the delta method [35].

Results were shown in point estimates and their 95% confidence intervals (CIs), reflecting beta coefficients for continuous variables or odds ratios (ORs) for binary outcomes. All analyses were performed using R 4.3.1 with the “TwoSampleMR” R package. Taking multiple testing into account, we applied Bonferroni correction in the MR analyses, thus, the threshold of statistical significance is $0.05/(\text{number of exposures} \times \text{number of outcomes})$. A nominal threshold of $p < 0.05$ but above the Bonferroni corrected significance threshold was considered as suggestive evidence for potential causal associations.

3 | Results

3.1 | The Total Effects of Lifestyles on Age-Related Diseases

UVMR analyses were used to investigate the causal effect of lifestyle factors on IgG N-glycosylation. In European populations, we assessed the causal associations between 32 lifestyles and 25 age-related diseases, and the results are provided in Figure 2A and Table S4. Conventional (for IVW MR) F -statistic values for individual instrument SNPs ranged from 29.78 to 3390.21 (Table S3), which suggested the absence of weak IVs. For East Asian populations, 13 lifestyle and 14 age-related diseases were adopted for MR analyses. The F -statistics of the screened SNPs were all >10 to avoid bias caused by weak IVs (Table S6). We found suggested evidence involving three lifestyle factors and ischemic stroke, namely, ever versus never drinker and osteoporosis, three lifestyle factors and lung cancer, drinks per week and colorectal cancer, milk consumption and prostate cancer, natto consumption and endometrial cancer, and three for gastric cancer (Figure 2B and Table S7). After multiple testing corrections, UVMR results were not statistically significant in both East Asians and Europeans, suggesting that the obtained causal associations are nominally meaningful and provide only suggestive evidence. The results of the pleiotropy and heterogeneity tests are shown in Tables S5 and S8.

3.2 | Effects of Lifestyles on IgG N-Glycosylation

The two-step MR design was applied to recognize the mediating role of IgG N-glycosylation between lifestyle and age-related diseases. In step one, we employed UVMR to evaluate the causal effect of lifestyles on IgG N-glycans. All variants were robustly associated with the exposure at F -statistics >10 (Tables S9 and S12). The study revealed four pairs of physical activity–glycan associations and 31 pairs of dietary habit–glycan associations in European populations using the threshold of $p < 0.05$ (Figure 3A and Table S10). The results of the pleiotropy test via UVMR

analyses of the associations between lifestyle factors and glycans are presented in Table S11. The IVW results revealed suggestive evidence of causal associations between seven lifestyle factors and glycans in East Asian populations, as described in Table S13 and Figure 3B. The evidence for the presence of horizontal pleiotropy in the MR-Egger regression analysis and MR-PRESSO is described in Table S14. However, after the application of multiple testing correction, no causality was observed in the UVMR design in both European populations ($p < 0.05/23 \times 32$ [i.e., 6.793×10^{-5}]) and East Asian populations ($p < 0.05/24 \times 13$ [i.e., 1.603×10^{-4}]).

Given the robust strength of genetic instruments (all F -statistics >10 , see Tables S15 and S18), we performed reverse MR analysis between IgG N-glycans and lifestyles to infer the direction of causal relationships. These analyses provided suggestive evidence of four pairs of IgG N-glycans–lifestyle associations, including moderate to vigorous physical activity, vigorous physical activity, and 49 dietary habits among European individuals (all $p < 0.05$; Table S16). After the application of multiple testing correction ($p < 0.05/23 \times 32$ [i.e., 6.793×10^{-5}]), in the European population, no causality was observed in reverse MR analysis. At least one sensitivity analysis confirmed these IVW estimates. In the reverse MR analyses among East Asian individuals, we identified 11 potential causal associations between direct glycan traits and lifestyles (Table S19). Following multiple testing correction, the reverse MR analysis revealed no significant causal relationships of lifestyles on IgG N-glycosylation in the East Asians using $p < 0.05/24 \times 13$ (i.e., 1.603×10^{-4}). Cochran’s Q test was used to assess potential heterogeneity, and we employed MR-Egger and MR-PRESSO to identify pleiotropy (refer to Tables S17 and S20 for more details).

3.3 | Effect of IgG N-Glycosylation on Age-Related Diseases

By MVMR analyses, we assessed the causal effect of potential mediators selected from step one on age-related diseases. Table S21 showed the results of MVMR analyses of lifestyles and IgG N-glycans associated with age-related diseases in individuals of European ancestry. MVMR analyses of 22 pathways did not provide evidence of direct effects between lifestyles and age-related diseases as the p -values were all well above 0.05. Similarly, no significant associations of IgG N-glycome with the pathways from poultry consumption to head and neck cancer were detected (Table S21). In addition, our MR analysis provided the suggestive evidence that genetically predicted GP22 was causally associated with cognitive function after considering dried fruit consumption (OR = 0.967, 95% CI = 0.944–0.991; $p = 0.008$; Table S21). The results of the MVMR analysis revealed the causal relationships of GP10 after Bonferroni correction using a p -value of $0.05/26$ (OR = 2.142, 95% CI = 1.575–2.913, $p = 1.195 \times 10^{-6}$) and GP15 (OR = 0.427, 95% CI = 0.296–0.615, $p = 4.802 \times 10^{-6}$) with RA (Table S21). In the MVMR analyses of fortified wine consumption and IgG glycans and AD, our findings revealed that GP24 could affect AD using the conventional threshold of 0.05 (OR = 1.050, 95% CI = 1.014–1.088; $p = 0.006$), and the associations of the remaining glycan traits were not significant. In East Asian populations, the association between IgG N-glycans and age-related diseases was no longer significant after controlling for the effect of lifestyles compared with the results of two-sample MR (all $ps > 0.05$; Table S22).

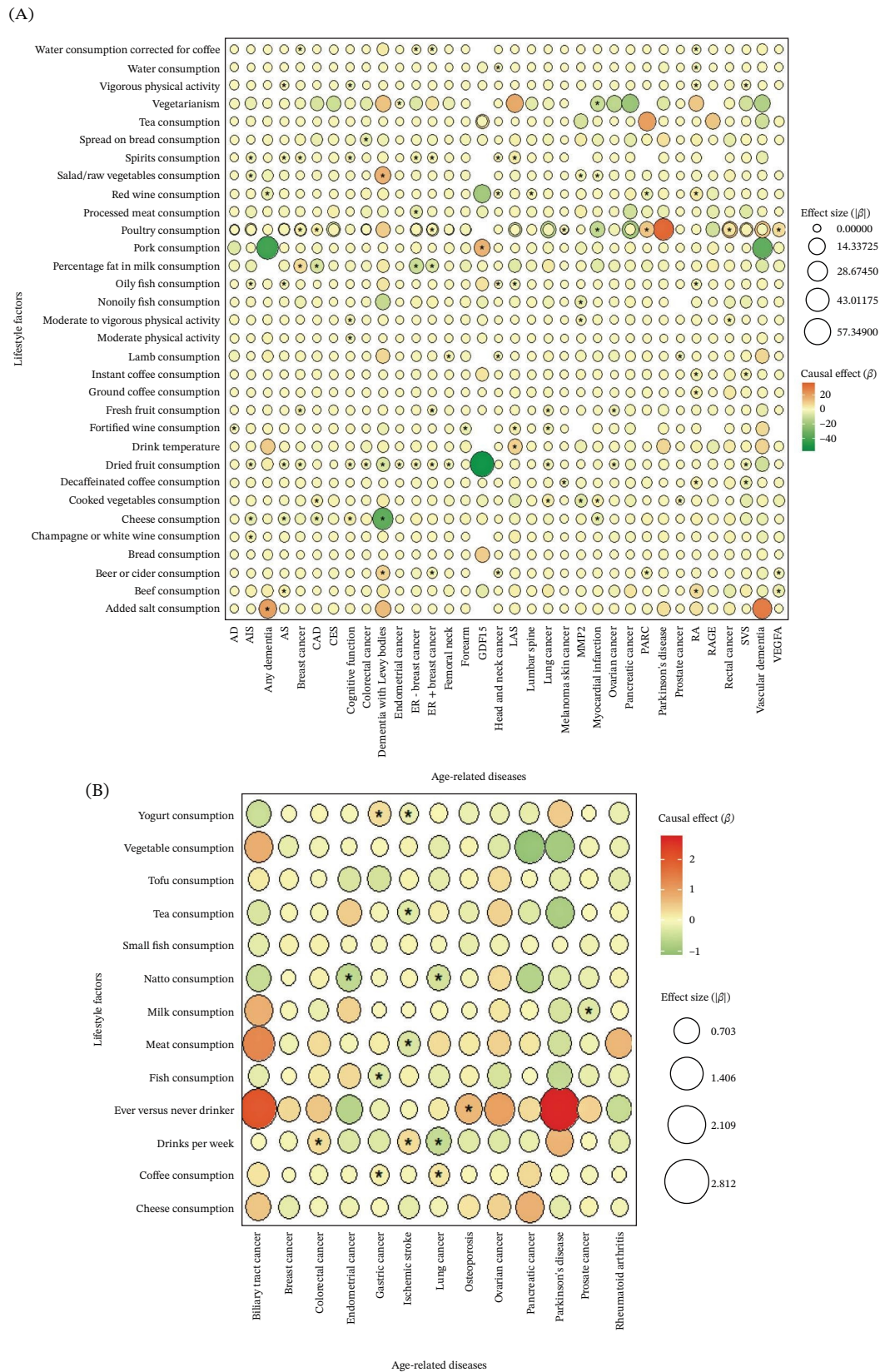
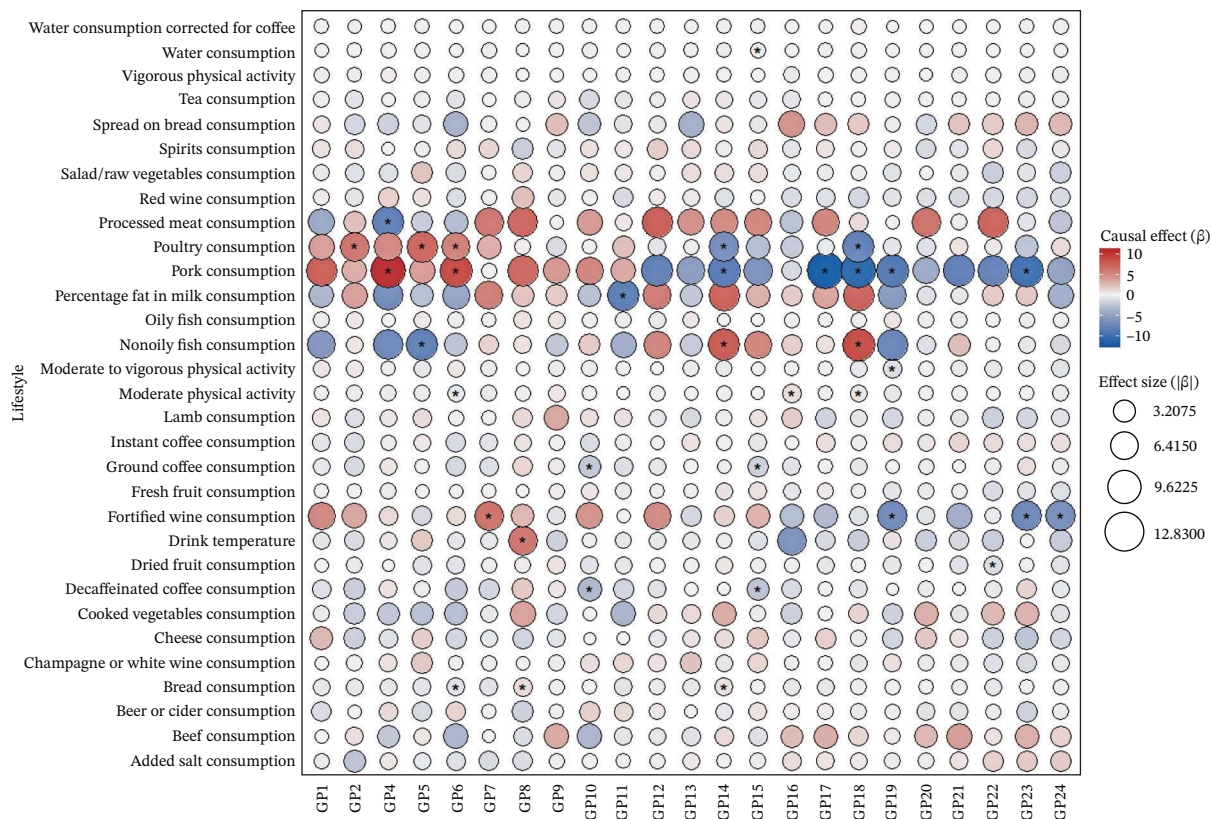


FIGURE 2 | The causal effects of lifestyles on age-related diseases. (A) In European populations. (B) In East Asian populations. *Significance markers represent $p < 0.05$.

(A)



(B)

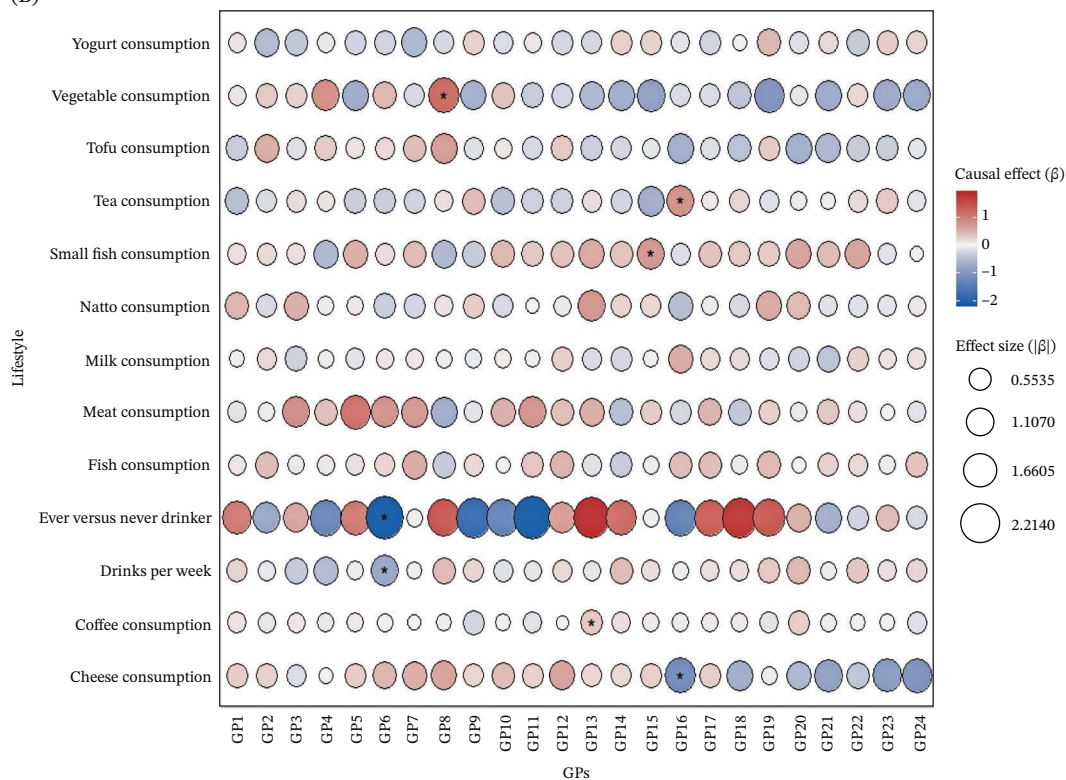


FIGURE 3 | The causal effects of lifestyles on IgG glycosylation. (A) In European populations. (B) In East Asian populations. GP, glycan peak. *Significance markers represent $p < 0.05$.

3.4 | Mediation Effect of IgG N-Glycosylation

The mediating effect estimation was conducted on three pathways with IgG N-glycans examined as potential mediators in individuals of European ancestry (i.e., dried fruit consumption → cognitive function, ground coffee consumption → RA, and fortified wine consumption → AD). The IgG N-glycans used in the analyses were causally associated with the above exposures and had causal effects on age-related diseases. However, there was no statistically significant evidence of GP22 as a mediator in the pathway from dried fruit consumption to cognitive function ($\beta = 0.063$; 95% CI = -0.005 to 0.073 ; $p = 0.092$; Figure 4A, and Table S23). In addition, our findings did not reveal that selected IgG N-glycans play a role in the causal effect of fortified wine consumption on AD, specifically, GP7 ($\beta = -0.062$; 95% CI = -0.082 to 0.059 ; $p = 0.315$), GP19 ($\beta = 0.415$, 95% CI = -0.148 to 0.978 ; $p = 0.149$), GP23 ($\beta = 0.140$, 95% CI = 0.095 – 0.374 ; $p = 0.244$) and GP24 ($\beta = -0.327$, 95% CI = -0.712 to 0.058 ; $p = 0.096$; Figure 4C, and Table S23). For the pathways associated with ground coffee consumption and RA, we examined the mediating effects of two potential mediators, namely, GP10 and GP15. Notably, at a nominal significance threshold of $p < 0.05$, our findings were suggestive of a potential mediating role of GP10 in the relationship between ground coffee consumption and RA, with an estimated suppression proportion of 70.4% ($\beta = -1.614$, 95% CI: -3.009 to -0.219 ; $p = 0.023$; Figure 4B and Table S23).

However, this association did not surpass the Bonferroni-corrected p -value threshold ($p < 0.05/7$). In contrast, no significant mediating was observed for GP15 ($\beta = 1.385$, 95% CI = -0.110 to 2.880 ; $p = 0.069$). One of the requirements for establishing mediating effects is the significant association between lifestyles and glycans. However, our findings revealed that there were no causal effects within East Asian populations, and we did not evaluate the mediating effect of glycan traits between lifestyles and age-related diseases.

4 | Discussion

To the best of our knowledge, this is the first investigation to evaluate the mediating effect of IgG N-glycosylation between lifestyles and age-related diseases in East Asian and European populations. By leveraging publicly available GWAS datasets of lifestyles, age-related diseases, and IgG N-glycans, this study employs UVMR, MVMR, and two-step MR to validate IgG N-glycans as mediators therein. Our findings might provide novel insights into how N-glycosylation links lifestyles to age-related diseases, advancing the understanding of this biological pathway.

Previous studies have investigated causal associations between physical activity and IgG N-glycosylation. Tijardović et al. [21] conducted a longitudinal exercise intervention among 29 male physical education students and collected their blood samples for

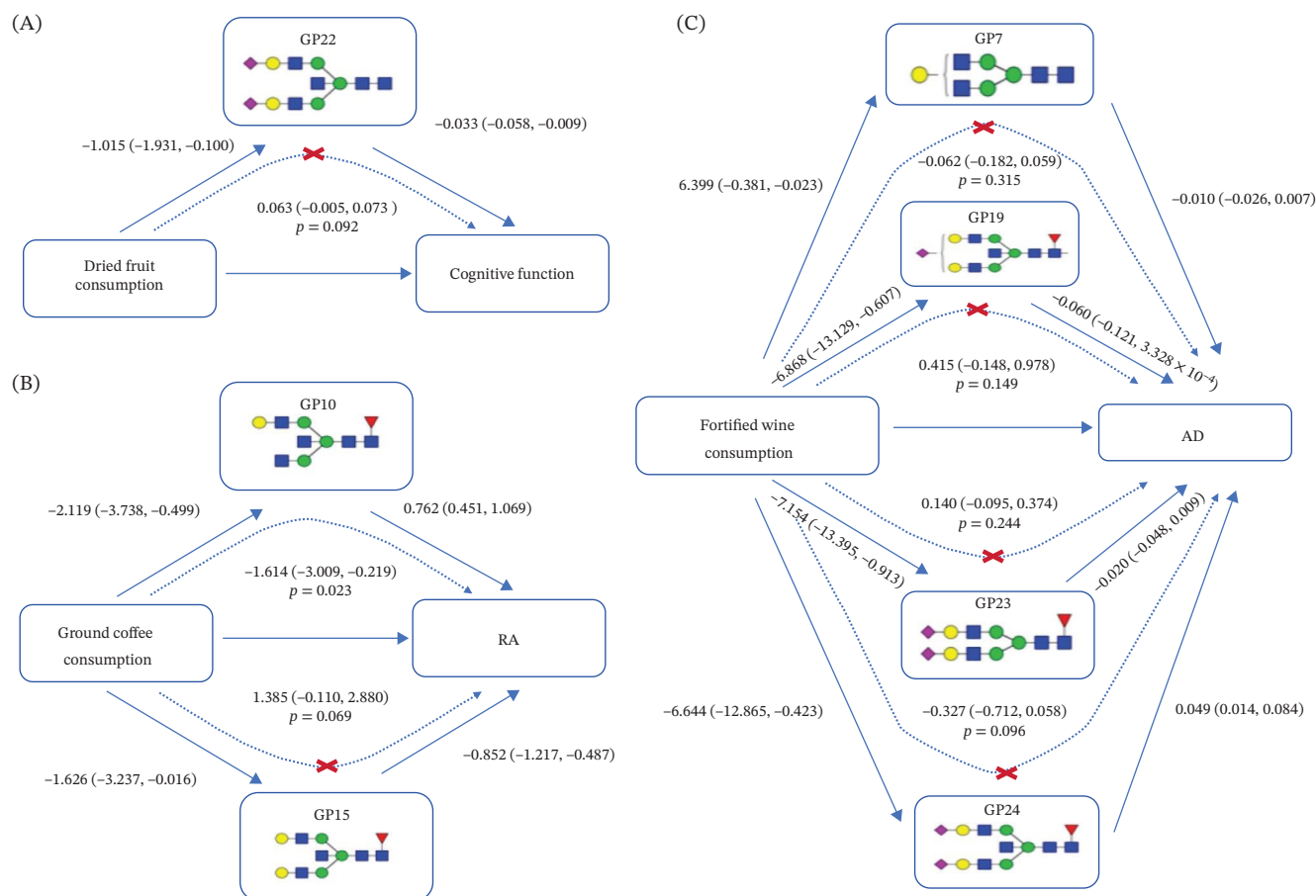


FIGURE 4 | The mediating effect of IgG glycosylation on the relationship between lifestyle and age-related diseases. GP, glycan peak; RA, rheumatoid arthritis. (A) The mediating role of GP22 linking dried fruit consumption with cognitive function. (B) The mediating roles of GP10 and GP15 linking ground coffee consumption with rheumatoid arthritis (RA). (C) The mediating roles of GP7, GP19, GP23, and GP24 linking fortified wine consumption with Alzheimer's disease (AD).

further N-glycosylation measurements. One month after the exercise intervention, which was not a complete exercise session, they observed a weak and significant change in the IgG N-glycome, which manifested as a decrease in the level of agalactosylated glycans and an increase in the level of digalactosylated glycans. In addition, a study performed in sedentary, middle-aged, overweight, and obese populations reported a significant increase in agalactosylated, monogalactosylated, asialylated, and core-fucosylated N-glycans and a decrease in digalactosylated, mono-, and disialylated N-glycans [36]. To the best of our knowledge, causal associations between IgG N-glycome and dietary habits have been less reported. The results of the aforementioned studies generally concur with those from our study. Furthermore, it has been widely reported that IgG N-glycosylation is causally associated with aging and age-related diseases [10, 37–40], as indicated by the decreased abundance of galactosylation and increased abundance of agalactosylation. Based on these observations, IgG N-glycosylation has

garnered significant attention in research on predicting biological age. Krištić et al. [11] and Yu et al. [12] built the glycan age utilizing age-related glycan traits in European and East Asian populations, respectively.

Using a two-step MR approach, we first applied bidirectional UVMR to assess the causal effects of lifestyles on IgG N-glycan traits (step one), followed by mediator selection based on predefined criteria. Only lifestyles-IgG N-glycan pairs with significant forward associations in UVMR were further analyzed to evaluate the mediating role of IgG N-glycans in age-related diseases (step two). Exposure-outcome pairs that attained significance in UVMR were carried forward into mediation MR analyses to evaluate the mediating effect of IgG N-glycans. (e.g., ground coffee consumption → RA, cheese consumption → stroke, poultry consumption → head and neck cancer; Figure 5A). While the effects of physical activity and IgG N-glycans have been identified, we did

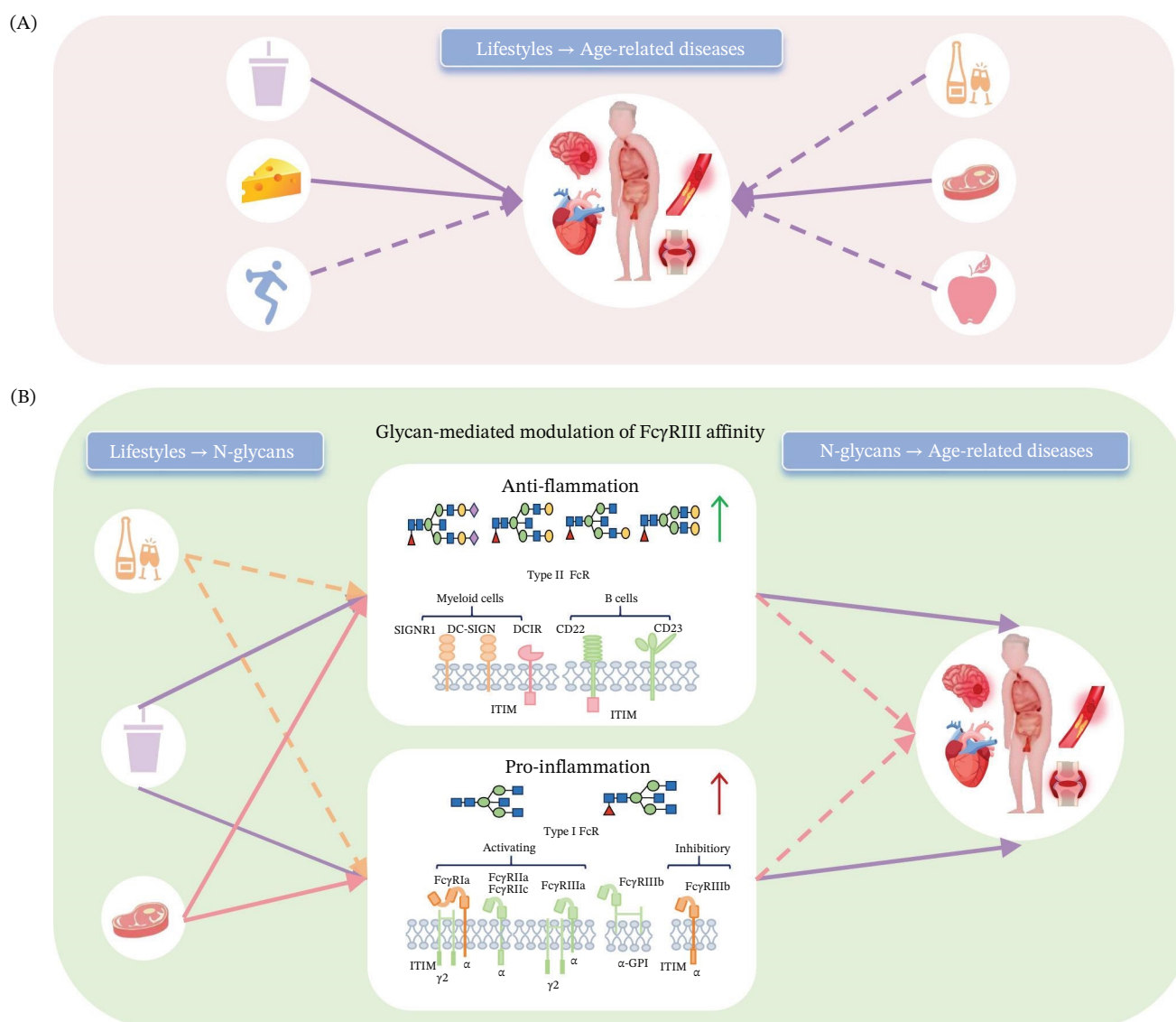


FIGURE 5 | The proposed mediating effects of IgG N-glycans between lifestyles and age-related diseases. DCIR: C-type lectin domain family 4 member A; DC-SIGN: CD209 antigen; Fuc: core fucose; Gal: galactose; GlcNAc: N-acetylglucosamine; ITAM: immunoreceptor-tyrosine-based activatory motif; ITIM: immunoreceptor-tyrosine-based inhibitory motif; SIGNR1: CD209 antigen-like protein B; Man: mannose; Neu5Ac: sialic acid; α -GPI: glycerolphosphatidylinositol. (A) The total effect of lifestyles on age-related diseases. (B) The proposed mediating effects of IgG N-glycans in the pathogenesis pathways linking lifestyles with age-related diseases.

not assess the mediating role of IgG N-glycans, as there were null associations between physical activity and several diseases, including AD. The potential regulatory effect of changes in N-glycans cannot be denied, and alterations in N-glycosylation, which encompasses galactosylation, sialylation, core-fucosylation, as well as bisecting GlcNAc, strongly influence the immune and inflammatory functions of IgG. As shown in Figure 5B, we did not conduct further MVMR analyses for champagne or white wine consumption since their null associations with IgG glycans. We also excluded pathways that showed null associations between IgG N-glycans and age-related diseases from MVMR analyses, and this exclusion aligns with the principle that variables lacking a significant causal effect in step two cannot serve as mediators (e.g., poultry consumption → head and neck cancer), as well as excluded complete mediation of IgG N-glycans (e.g., decaffeinated coffee consumption → SVS).

In our study, we found three pathways that N-glycans might play as mediators, namely, GP22 as a potential mediator in the pathway from dried fruit consumption and cognitive function; GP10 and GP15 as potential mediators in the pathway from ground coffee consumption and RA; and GP7, GP19, GP23, and GP24 as potential mediators in the pathway from fortified wine consumption and AD. However, we did not observe the significant mediating effect of IgG glycans between dried fruit consumption and cognitive function and between fortified wine consumption and AD. Intriguingly, our findings revealed the negative mediating effect of GP10 (FA2 [6]BG1) in the pathway from ground coffee consumption to RA. Our finding is biologically plausible. Oxidative stress can trigger the activation of diverse transcription factors that prompt the initiation of inflammation [41], whereas coffee consumption has been found to reduce biomarkers of oxidative stress and inflammation due to its abundant antioxidant properties. GP10 consists of monogalactosylated and fucosylated structures with bisecting GlcNAc. Previous studies have indicated that galactosylation of IgG glycans may contribute to the regulation of inflammatory responses via interactions with FcγRIIB and Dectin-1, thereby inhibiting complement-mediated inflammation [42]. Moreover, bisecting GlcNAc modification can affect terminal glycosylation processes and suppress the biosynthesis of core fucose and sialic acid [43, 44]. In addition, coffee consumption may reduce oxidative stress and inflammatory markers, providing a plausible biological context for the observed association [41]. The SMARCB1-DERL3 locus has been linked to the regulation of GP10 levels, supporting a potential molecular link between IgG N-glycosylation and RA pathogenesis [45]. Notably, IgG N-glycosylation mediates diverse immunological functions that are influenced by multiple factors, including IgG subclass, glycosylation patterns, receptor binding, disease context, and cell-specific immune responses. The mediating effect of GP10 on the association between coffee consumption and RA was generated from hypothesis-driven MR analysis. These findings should be considered suggestive evidence rather than confirmed causality, and they require independent validation through mechanistic experiments and prospective cohort studies. Moreover, another finding that needs to be paid attention to was the bidirectional causal associations of IgG N-glycome and SASP that were reported in our previous work [46], namely, the associations of the mediator with the outcomes were not unidirectional. It is worth noting that the direction of mediation-outcome was required to be consistent with the direction of exposure-

mediator in mediation MR studies [47, 48], rather than bidirectional. The roles of IgG N-glycome link lifestyles with aging, as well as age-related, are worth further exploration.

The present study is novel in applying mediation MR to evaluate IgG N-glycosylation as a potential mediator between lifestyle factors and age-related diseases, offering suggestive evidence that merits further exploration in terms of underlying mechanisms. This approach provides a powerful framework for causality, advancing beyond conventional observational studies by addressing confounding and reverse causation. We also emphasize the need for more powerful functional studies of IgG N-glycosylation to further elucidate the underlying molecular mechanisms and to validate its mediating role in the development and progression of age-related diseases. Our findings may inform future research into the biological pathways linking IgG glycosylation to inflammation and age-related diseases. However, these potential implications are speculative and require substantial experimental validation before any clinical or therapeutic relevance can be considered.

Although we did not observe significant mediating effects of IgG N-glycome between lifestyles and age-related diseases in East Asians, this null finding must be interpreted with caution. Given the relatively small sample size ($n = 536$) of East Asians, the overall mediating effects of IgG N-glycans between lifestyles and age-related diseases may be due to the true effect being too modest to be detected. Additionally, significant heterogeneity in dietary exposures existed between the Chinese and Japanese, reflecting inherent differences in the source populations from which the data were derived (Xuanwu Hospital, China, and BBJ, respectively). This cross-population heterogeneity may introduce a potential influence on qualifying the mediating effects of IgG N-glycans to some extent. Our findings support the role of GP10 as a potential mediator in the pathway linking ground coffee consumption to RA in European populations, though this mediating effect was not observed in East Asians, which might be due to ethnic-specific differences in the IgG glycosylation pattern, explained by both genetic and environmental factors [49].

This study performed MR analyses to explore the potential mediating roles of IgG N-glycosylation in the pathway from lifestyles to aging and aging-related diseases and provides evidence illustrating that IgG N-glycosylation was qualified as a mediator therein. The main strengths of the present study lie in its MR study design, which incorporates normative and specific criteria for mediator screening, along with the robust causal associations from the main analysis, further validated by sensitivity analyses. Despite the fact that the MR design takes advantage of genetic variants as IVs to detect causal associations, which are less prone to confounding than observational studies, limitations must be declared. First, considering that genetic variants usually account for only a small proportion of the variability in phenotype, MR requires a large sample size. Although we obtained the largest summary-level data of IgG N-glycans ($N = 8090$ in European populations and $N = 536$ in East Asian populations), the sample size was still limited in comparison with the exposures and outcomes used in this study. Further studies with a large population are needed. Second, the present study takes direct IgG N-glycan traits that present significant associations in step one into account in MVMR analyses, and the interaction of multiple

mediators cannot be modeled in the present two-step MR framework; in particular, IgG N-glycan traits share the same structure. In addition, while pleiotropy was evaluated through MR-Egger and MR-PRESSO analyses, it remains impossible to entirely rule out the potential for residual pleiotropic effects. Last, the potential effects of alterations in Fab glycosylation can be nonnegligible, especially for the relatively small sample size ($n = 536$) of East Asians. Nevertheless, the measurement of glycosylation detected was the HILIC-UPLC method, which cannot distinguish the difference between Fab glycosylation and Fc glycosylation. Given that only 15%–20% of the Fab region is glycosylated, the causal relationships between IgG N-glycans and aging-related disease, as well as lifestyle factors, are predominantly influenced by Fc glycans. Nonetheless, the potential contribution of Fab N-glycosylation cannot be fully excluded; residual pleiotropy may remain.

5 | Conclusion

In summary, the mediation MR design provides suggestive evidence that GP10 may play a role in the association between coffee consumption and RA. However, these findings should be interpreted as hypothesis-generating from MR rather than as causality. Further independent validation through mechanistic studies and prospective cohorts is warranted to test the observed mediation pattern and to explore potential underlying pathways.

Author Contributions

Conceptualization: Youxin Wang, Jinlong Li, Jingyu Wang, and Haotian Wang. Data curation: Haotian Wang, Jingyu Wang, and Xiaoni Meng. Formal analysis: Haotian Wang and Jingyu Wang. Funding acquisition: Youxin Wang. Investigation: Haotian Wang, Jingyu Wang, Xiaoni Meng, Shengzhi Sun, and Youxin Wang. Methodology: Haotian Wang, Jingyu Wang, Xiaoni Meng, and Youxin Wang. Supervision: Shengzhi Sun and Youxin Wang. Validation, visualization: Haotian Wang. Writing – original draft preparation: Haotian Wang and Jingyu Wang. Writing – review and editing: Haotian Wang, Jingyu Wang, Xiaoni Meng, Shengzhi Sun, Jinlong Li, and Youxin Wang.

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Ethics Statement

The ethical approvals of IgG N-glycosylation involved in the East Asian population were approved by the Ethics Committee of Capital Medical University (Beijing, China) with approve number 2009SY16, and written informed consent was obtained from each subject at the beginning of the study. In addition, the ethical approval of the remaining phenotypes was not sought since we obtained summary-level data from publicly available datasets from GWASs, and no individual-level data were included.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets generated during and/or analyzed during the current study are available from the repositories listed in Table S1 and additional files.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Supporting Information.** STROBE Statement—Checklist of items that should be included in reports of Mendelian randomization. Table S1: Basic information of summary statistics data sources in the MR analyses. Table S2: The structures and descriptions of measured IgG N-glycan traits. Table S3: Summary of F-statistics for genetic instrument strength across lifestyles and age-related diseases in Europeans. Table S4: Total effect of lifestyles on age-related diseases in European populations. Table S5: Horizontal pleiotropy and heterogeneity

tests of lifestyles with age-related diseases in Europeans. Table S6: Summary of F-statistics for genetic instrument strength across lifestyles and age-related diseases in East Asians. Table S7: Total effect of lifestyles on age-related diseases in East Asian populations. Table S8: Horizontal pleiotropy and heterogeneity tests of lifestyles with age-related diseases in East Asians. Table S9: Summary of F-statistics for genetic instrument strength across lifestyles and IgG N-glycans in Europeans. Table S10: The causal associations of lifestyles on IgG N-glycans in European populations. Table S11: Horizontal pleiotropy and heterogeneity tests of lifestyles with IgG N-glycans in Europeans. Table S12: Summary of F-statistics for genetic instrument strength across lifestyles and IgG N-glycans in East Asians. Table S13: The causal associations of lifestyles on IgG N-glycans in East Asian populations. Table S14: Horizontal pleiotropy and heterogeneity tests of lifestyles with IgG N-glycans in East Asians. Table S15: Summary of F-statistics for genetic instrument strength across IgG N-glycans and lifestyles in Europeans. Table S16: The causal associations of IgG N-glycans on lifestyles in European populations. Table S17: Horizontal pleiotropy and heterogeneity tests of IgG N-glycans with lifestyles in Europeans. Table S18: Summary of F-statistics for genetic instrument strength across IgG N-glycans and lifestyles in East Asians. Table S19: The causal associations of IgG N-glycans on lifestyles in East Asian populations. Table S20: Horizontal pleiotropy and heterogeneity tests of IgG N-glycans with lifestyles in East Asians. Table S21: Multivariable MR estimates of the effect of IgG N-glycans on age-related disease after adjusting for exposures in Europeans. Table S22: Multivariable MR estimates of the effect of IgG N-glycans on age-related disease after adjusting for exposures in East Asians. Table S23: The detailed mediation proportions of mediators.